

S4G Image Workflow Documentation

Overview

The workflow downloads S4G FITS images, builds a geometry manifest, and plots diagnostic isophote PDFs.

Workflow:

1. download_s4g_images.py

Downloads each galaxy's S4G 3.6 micron FITS image from IRSA.

Output: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\s4g_images_36um

2. build_s4g_geometry_manifest.py

Combines the downloaded FITS image paths with local and catalogue geometry data.

Output: s4g_image_downloader\geometry_output\s4g_image_geometry_manifest.csv

3. plot_s4g_isophote_axes.py

Reads the manifest and FITS images, then creates major/minor-axis isophote diagnostic PDFs.

Output: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\isophote_output

Function-level overview: download_s4g_images.py

- parse_args: Defines command-line options such as --limit, --dry-run, --outdir, and --timeout.
- read_galaxy_names: Reads scrambled_map.txt and returns the galaxy names from the third column.
- download_image: Builds the IRSA FITS URL, skips existing files, optionally dry-runs, and saves valid downloads.
- main: Resolves paths, creates the output directory, reads the galaxy list, and downloads each requested image.

Function-level overview: build_s4g_geometry_manifest.py

- clean_value: Converts table text to numbers and normalizes known missing or invalid values to None.
- read_scrambled_map: Loads scrambled index, original index, and galaxy name from scrambled_map.txt.
- read_s4g_table: Reads local s4gbars_table.dat and keeps cleaned S4G bar and galaxy fields by name.
- table_value: Converts Astropy masked, scalar, or byte values into ordinary Python values.
- read_cached_or_fetch_table: Loads a VizieR catalogue from cache or downloads it and writes an ECSV cache file.

- `keyed_table`: Turns a catalogue table into a dictionary keyed by a selected name column.
- `keyed_herrera_bars`: Filters Herrera-Endoqui rows to bar entries and keys them by galaxy name.
- `fetch_catalog_geometry`: Fetches or loads Herrera-Endoqui, Salo, and Diaz-Garcia geometry catalogues.
- `fits_metadata`: Reads FITS header metadata needed for image size, pixel scale, centre, and orientation.
- `build_manifest`: Creates one manifest row per galaxy, merging local tables, catalogue geometry, FITS metadata, and notes.
- `write_manifest`: Writes the manifest rows to CSV using the fixed `OUTPUT_FIELDS` column order.
- `parse_args`: Defines paths and catalogue-cache switches such as `--image-dir`, `--no-vizier`, and `--refresh-cache`.
- `main`: Loads optional catalogue geometry, builds the manifest, writes the CSV, and prints image-link counts.

Function-level overview: `plot_s4g_isophote_axes.py`

- `parse_float`: Safely converts manifest string values to finite floats.
- `safe_filename`: Creates filesystem-safe PDF names from galaxy names.
- `read_manifest`: Reads the geometry manifest CSV into dictionaries.
- `pa_endpoint`: Converts a position angle and radius into x/y endpoints for drawing or sampling lines.
- `draw_pa_line`: Draws major or minor axis lines on the image panel.
- `extract_centered_subimage`: Crops a centre-based image cutout and builds arcsecond coordinate arrays.
- `profile_at_pa`: Samples a smoothed intensity profile along a requested position angle.
- `robust_log_image`: Creates a stable log-scaled image and contour levels even when pixels include zeros or NaNs.
- `deprojected_profile_radius`: Uses `angle_utils.deprojectr` to convert projected profile radii to deprojected radii.
- `required_geometry`: Pulls required geometry fields from a manifest row and rejects incomplete rows.
- `make_plot`: Creates the two-panel diagnostic plot for a single galaxy and writes it to individual and/or combined PDFs.
- `selected_rows`: Applies `--names` and `--limit` filters to the manifest rows.
- `parse_args`: Defines plotting options such as `--manifest`, `--image-dir`, `--output-dir`, `--names`, and `--no-combined`.
- `main`: Creates output folders, iterates through selected galaxies, writes PDFs, and reports

skipped cases.

Program detail: download_s4g_images.py

Default output: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\s4g_images_36um

Key commands: `python download_s4g_images.py --dry-run --limit 5`; `python download_s4g_images.py`

Program detail: build_s4g_geometry_manifest.py

Default output: s4g_image_downloader\geometry_output\s4g_image_geometry_manifest.csv

Key commands: `python build_s4g_geometry_manifest.py`; `python build_s4g_geometry_manifest.py --no-vizier`

Program detail: plot_s4g_isophote_axes.py

Default output: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\isophote_output

Key commands: `python plot_s4g_isophote_axes.py`; `python plot_s4g_isophote_axes.py --limit 3`

Recommended run order:

1. `python download_s4g_images.py --dry-run --limit 5`
2. `python download_s4g_images.py --limit 5`
3. `python build_s4g_geometry_manifest.py --no-vizier`
4. `python plot_s4g_isophote_axes.py --limit 3`